

Musculoskeletal System

Bone and Cartilage

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Objectives

1. Describe the normal function and structure of bone.
2. Describe the normal function and structure of cartilage.
3. Describe osteogenesis as well as bone repair strategies.
4. Describe regulation for bone remodeling.
5. Describe markers for bone disorders

Outline

- 1- Cartilage structure and function
- 2- Bone structure and function.
- 3- Cellular components.
- 4- Osteogenesis.
- 5- Hormonal regulation for calcium levels
- 6- Bone remodeling and repair.
- 7- Bone vs Dentin vs Enamel

Musculoskeletal system

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graph TD; A[Musculoskeletal system] --> B[Muscular System]; A --> C[Skeletal System];
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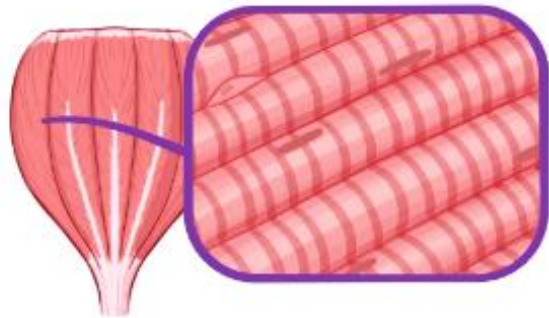
Muscular System

Skeletal System

Muscular System

* MADE of CONTRACTILE CELLS (MUSCLE FIBERS)

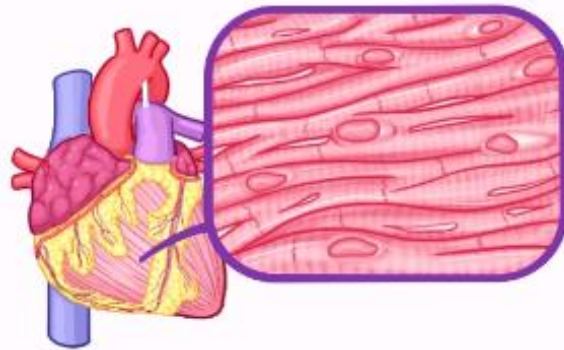
SKELETAL



- * CONNECT to the SKELETON & OTHER STRUCTURES (e.g. eyes)
- * HELP w/ MOVEMENT & STABILITY
- * VOLUNTARY

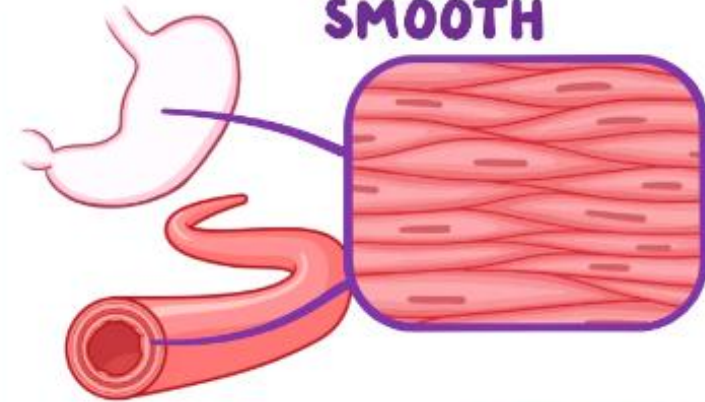


CARDIAC



- * WALLS of HEART
- * CONTRACT in RHYTHMIC WAY to PUMP BLOOD
- * INVOLUNTARY

SMOOTH

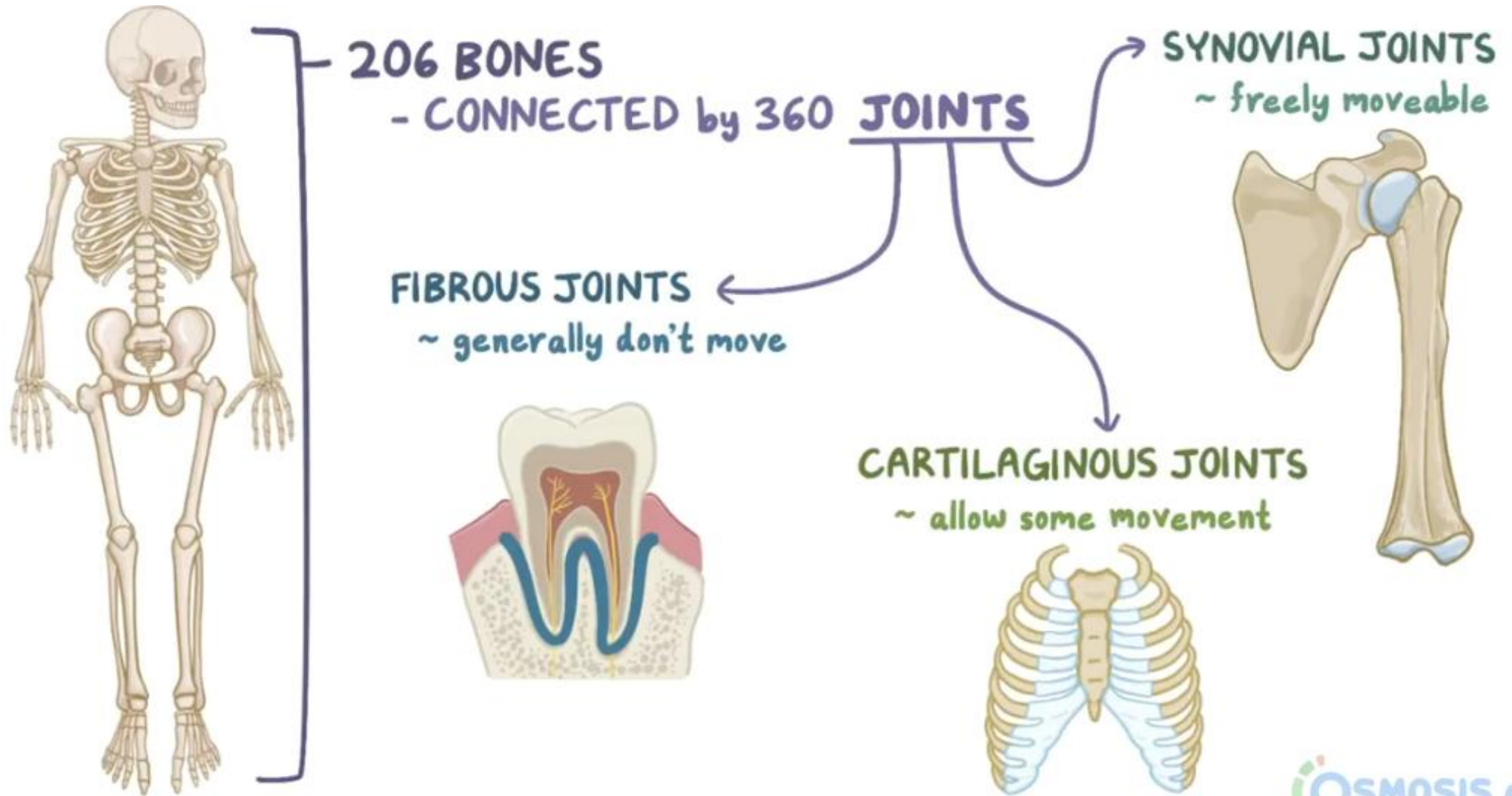


- * WALLS of:
 - ~ BLOOD VESSELS
 - └ contract walls
 - └ control blood flow
 - ~ HOLLOW ORGANS
 - └ peristaltic contractions
 - └ move contents of organs in one direction
- * INVOLUNTARY



OSMOSIS.or

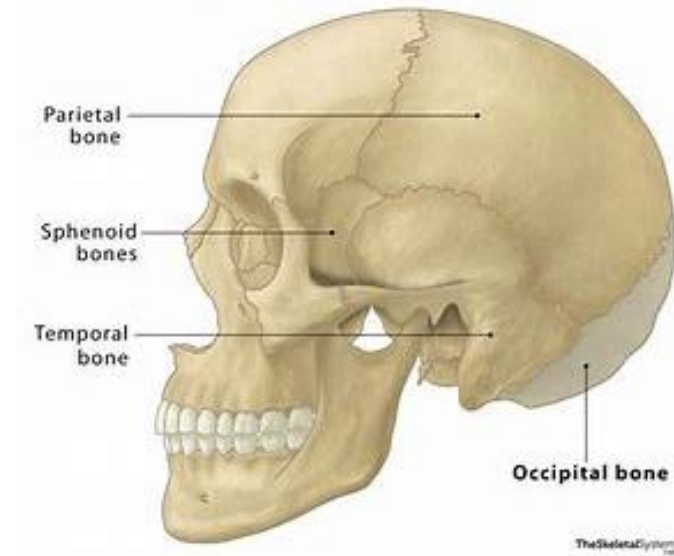
Skeletal System



Axial and appendicular skeleton

Embryonic Development

- All the bones in the appendicular and most axial skeleton: from mesoderm.
- Exception: skull bones. Facial bones and rostral skull bones develop from neural crest cells (----- origin).
- Neural crest cells develop into multipotent ectomesenchyme cells that develop into different bones (like maxilla, mandibles, frontal skull bones).
- Cartilage develops from mesoderm layer via chondrogenesis (mesenchyme).

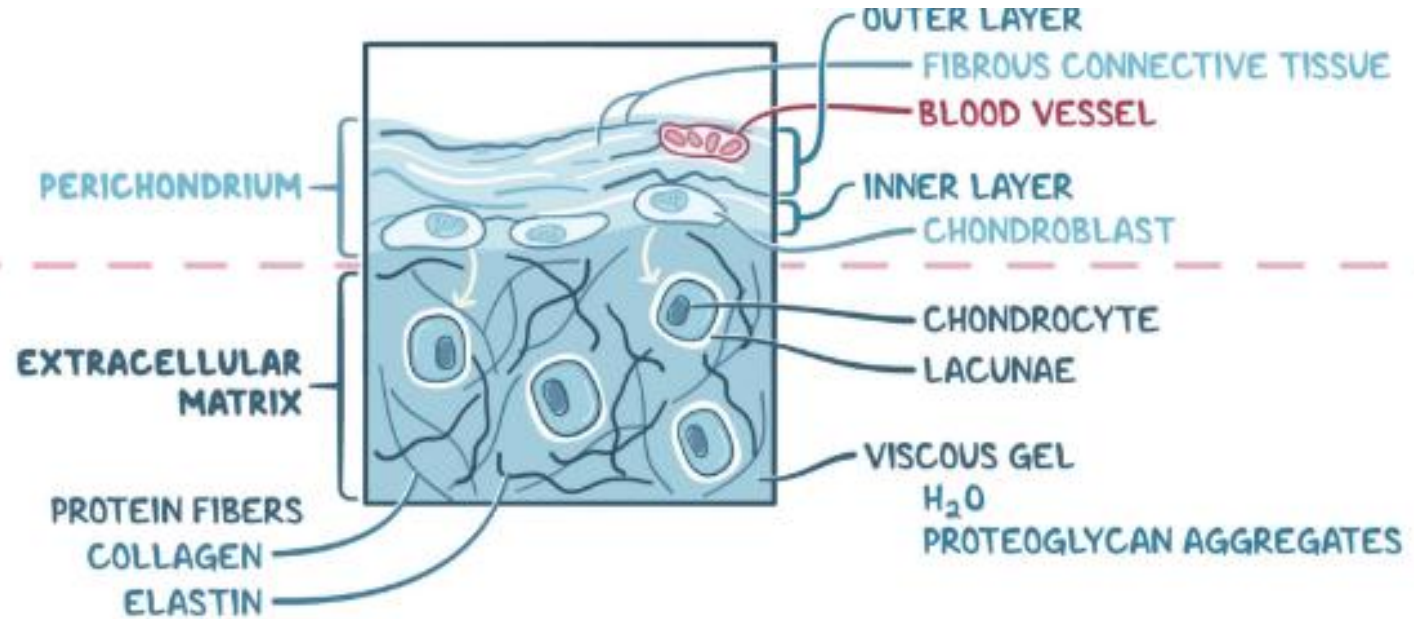


Histology and physiology

1- Cartilage

- Trough, flexible connective tissue.
- Structurally support soft tissues and provide cushioned, low-friction surface within joints.
- Chondrogenesis: embryonic development of cartilage from the mesenchyme.
- Cartilage is always avascular tissue that also lack lymphatic vessels and nerves. Gets nutrients from the perichondrium.

Cartilage =
chondrocytes + chondroblasts,
embedded in the extracellular
matrix (ECM).

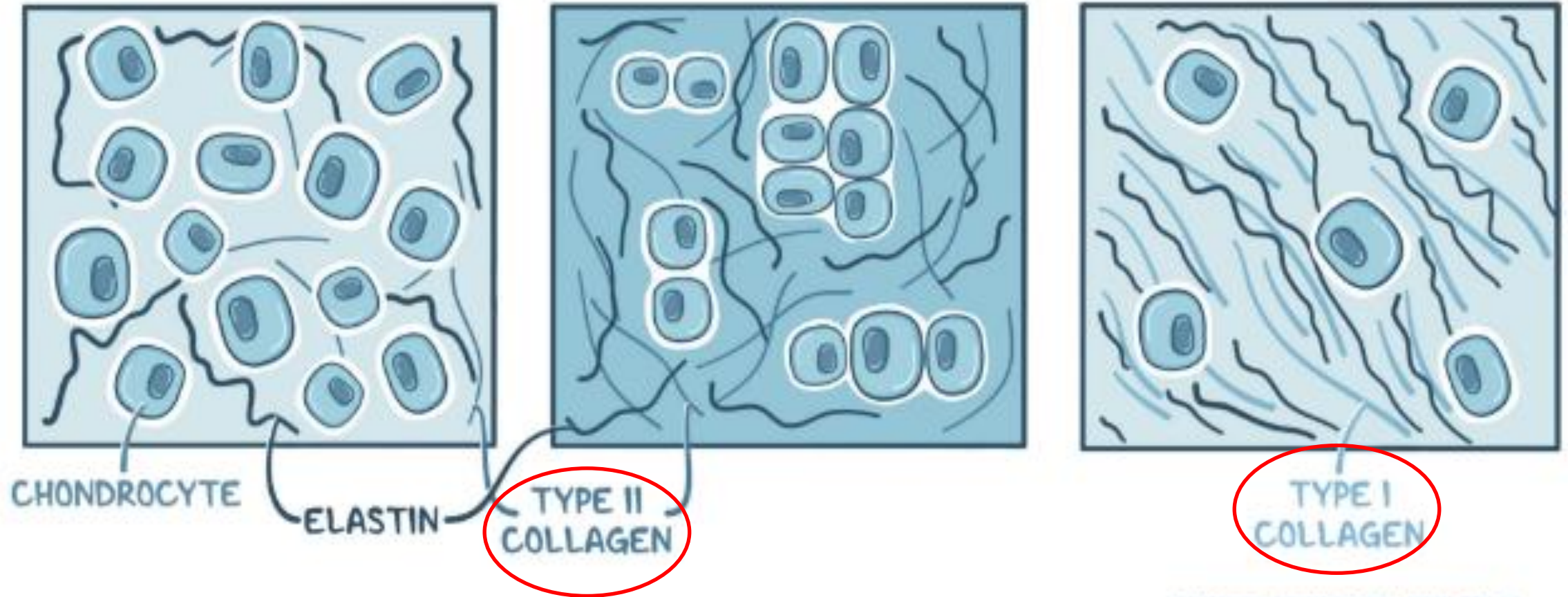


- The ECM includes collagen and proteoglycans that can bind a large amount of water.
- Chondroblasts undergo mitosis, secrete ECM components and mature into nondividing -----.
- Chondrocytes produce the small amount of ECM and maintain the cartilage long-term.

ELASTIC CARTILAGE

HYALINE CARTILAGE

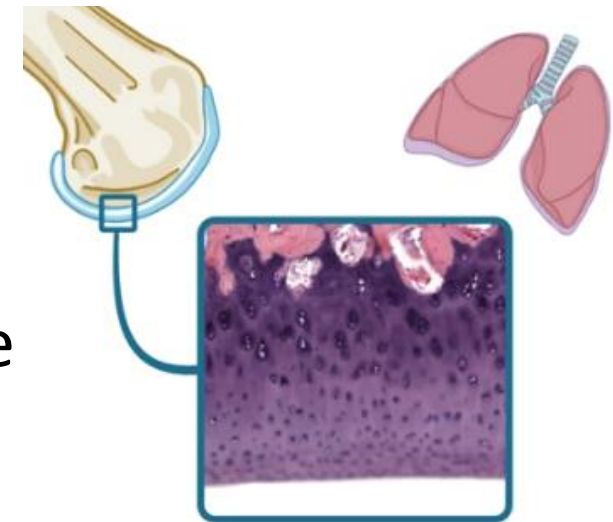
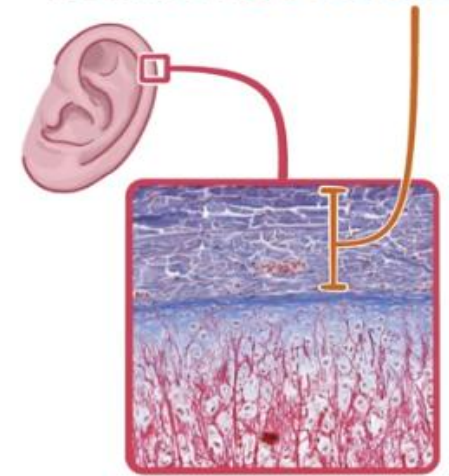
FIBROCARILAGE



- **Elastic cartilage:** least common but most elastic type. Has an abundant network of elastic fibers type II collagen.
- **Hyaline cartilage:** Most common type. Medium chondrocyte density. Found: in articular surfaces of movable joints, respiratory passages and rib ends.
 - In the embryo, hyaline cartilage forms the temporary skeleton later gradually replaced by bone.
 - Osteoarthritis involves the gradual loss or changed physical properties of the hyaline cartilage that lines the joints.

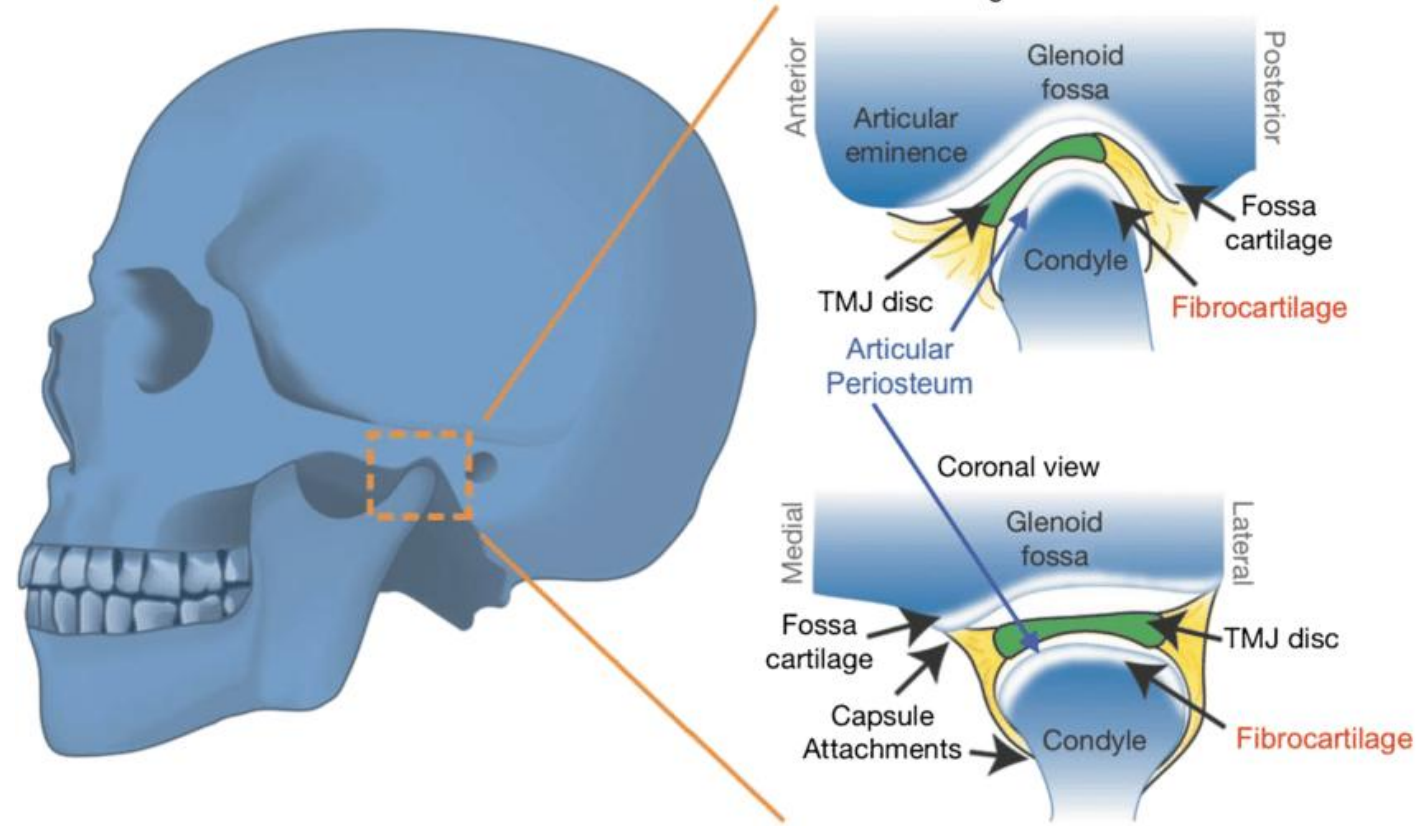
ELASTIC CARTILAGE

- ~ PINNA & EPIGLOTTIS
- ~ DENSE NETWORK of BRANCHING ELASTIC FIBERS
- ~ ALWAYS HAS a PERICHONDRUM



HYALINE CARTILAGE

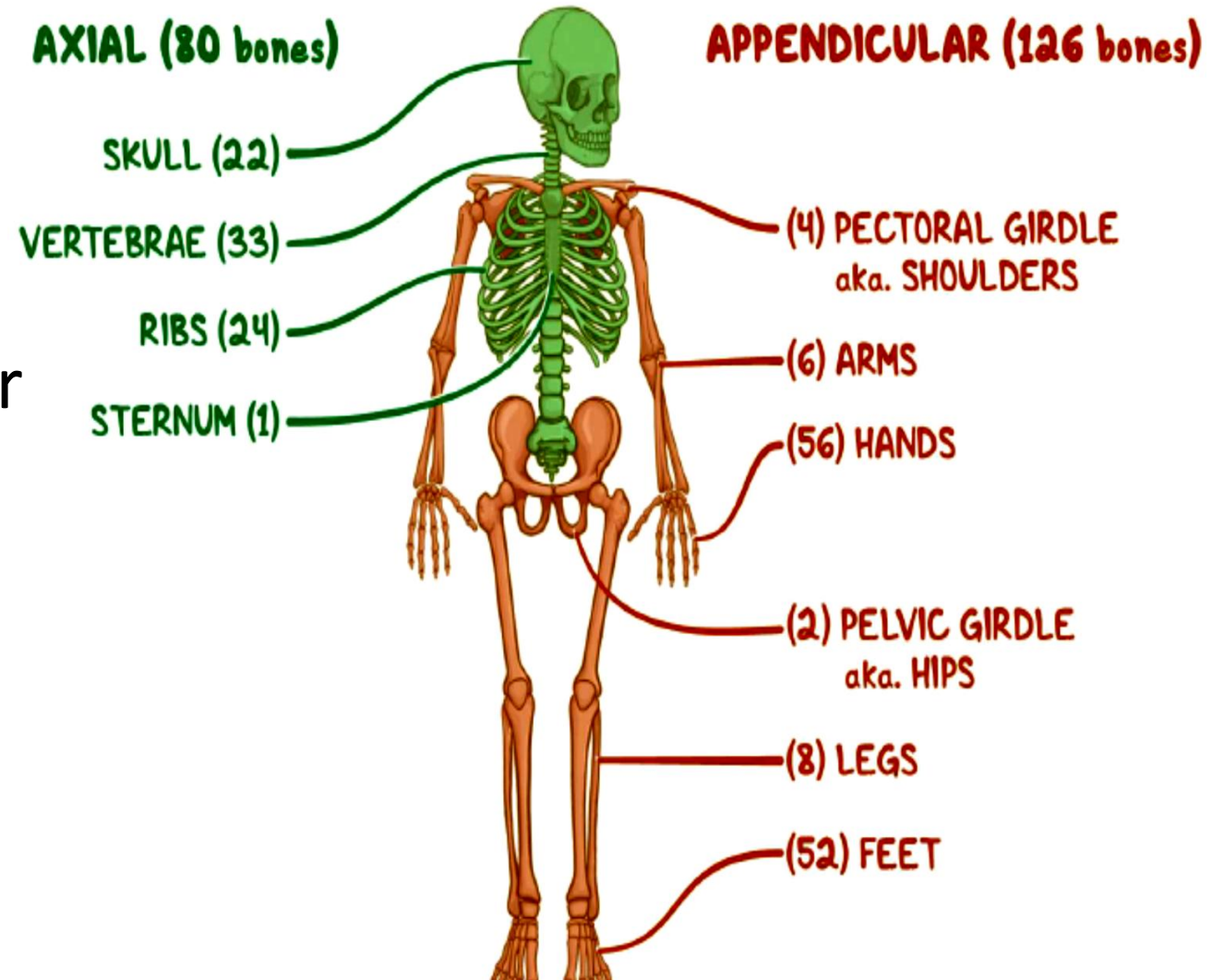
Fibrocartilage: common type. Highest tensile strength, least chondrocyte density but highest protein fiber density (type I collagen). Found meniscus of knee, spinal intervertebral discs, tendon insertion (e.g TMJ). Provides tough and cushioning support for the bone.



2- Bone

Bone is supportive connective tissue with calcified extracellular matrix.

Type I collagen matrix provides network for hydroxyapatite crystals to adhere.



Bone Function

Mechanical support: the skeleton provides support for the soft tissue and point of attachment for muscles.

Protective function: The skeleton protects the vital organs.

Bones maintain calcium and phosphate homeostasis.

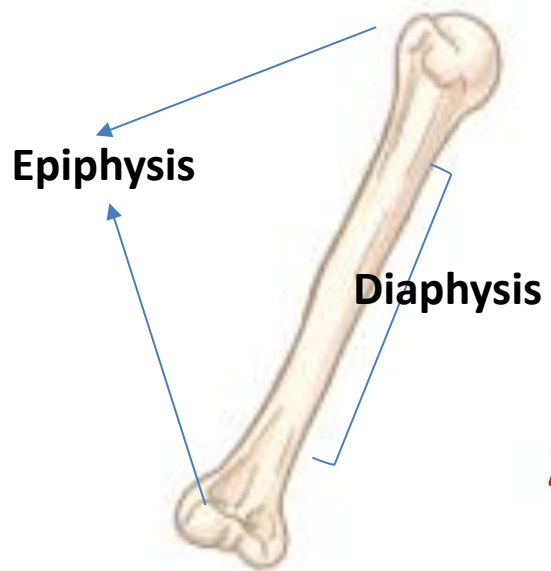
Bones enclose the bone marrow which is essential for blood cells formation.

In the ear, bones transduce sound waves from the ear drum to the inner ear.

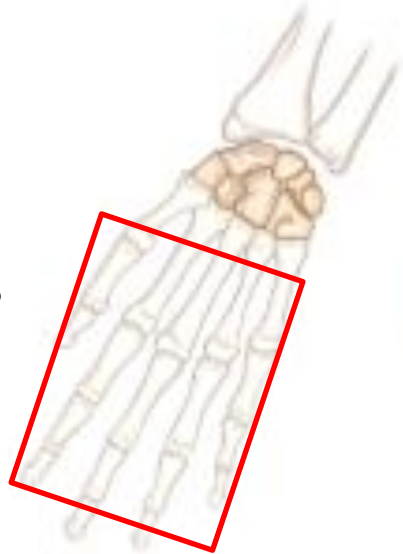
Aid in fatty acid storage in the form of yellow marrow.

Types of Bones

BONES CLASSIFIED by SHAPE



LONG



SHORT



FLAT

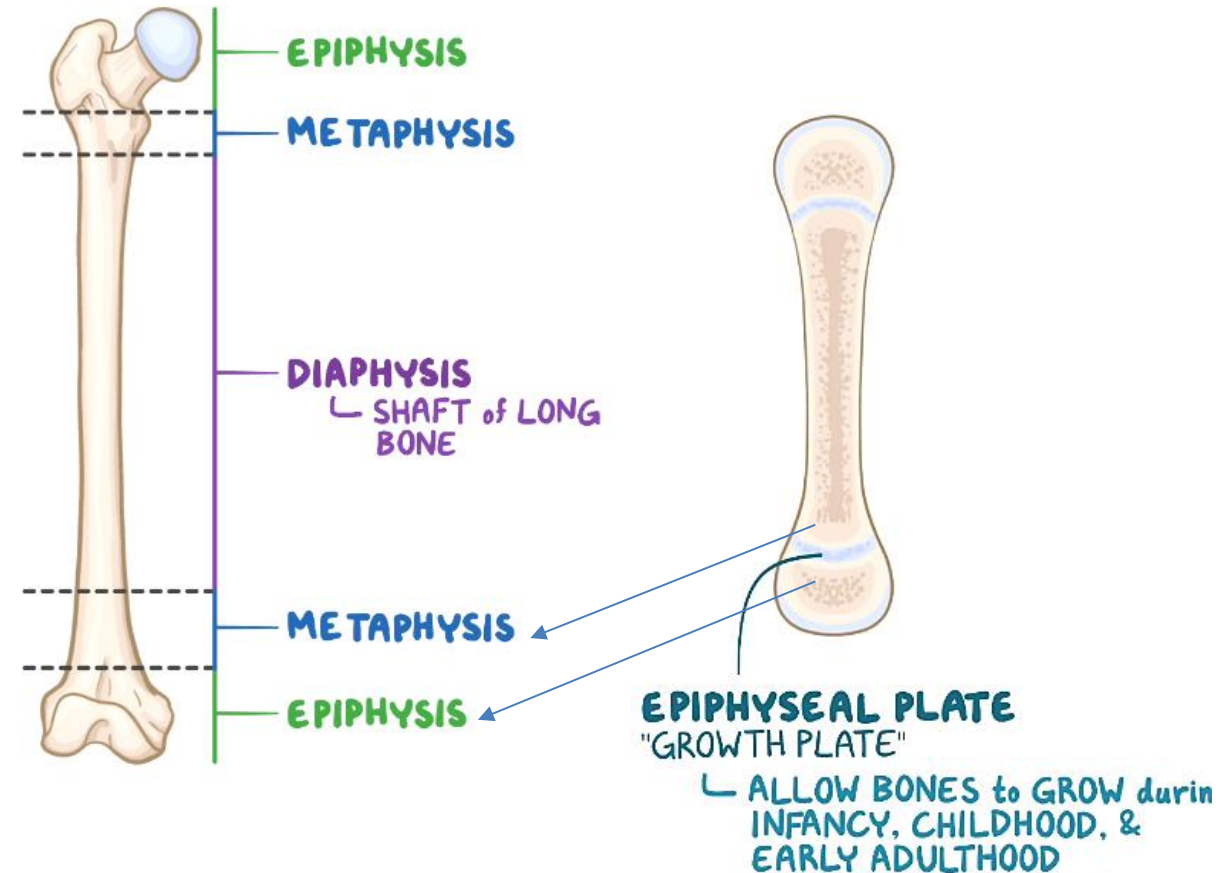
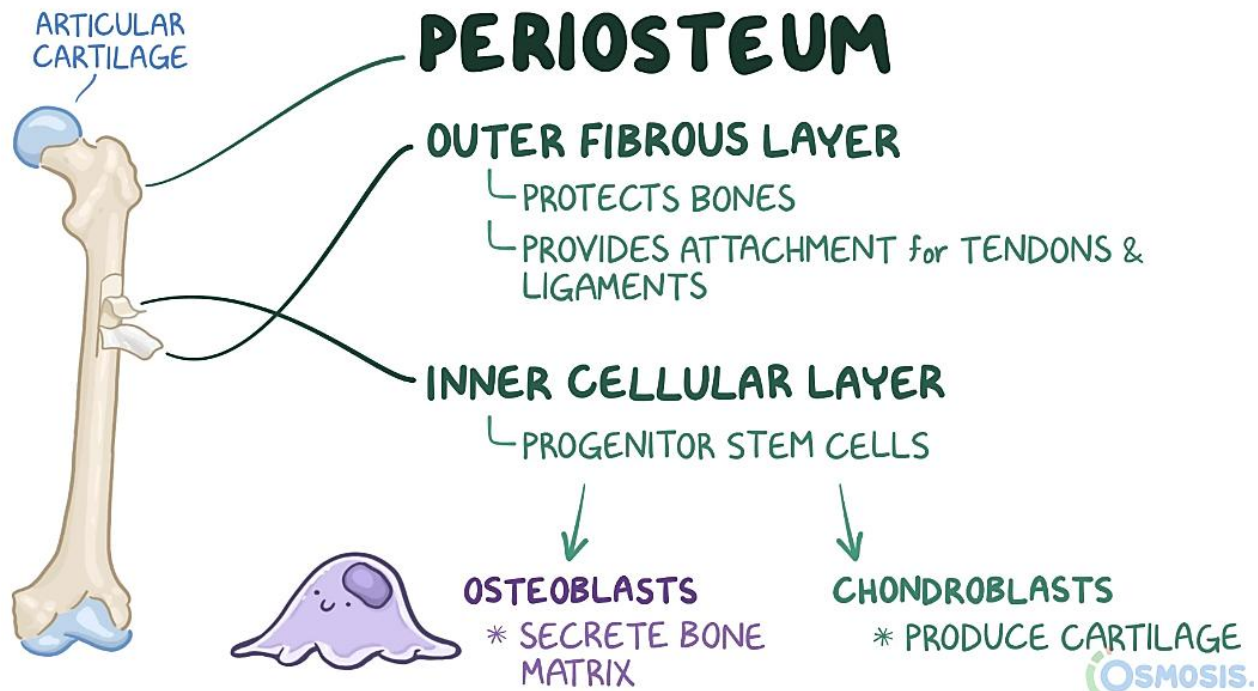


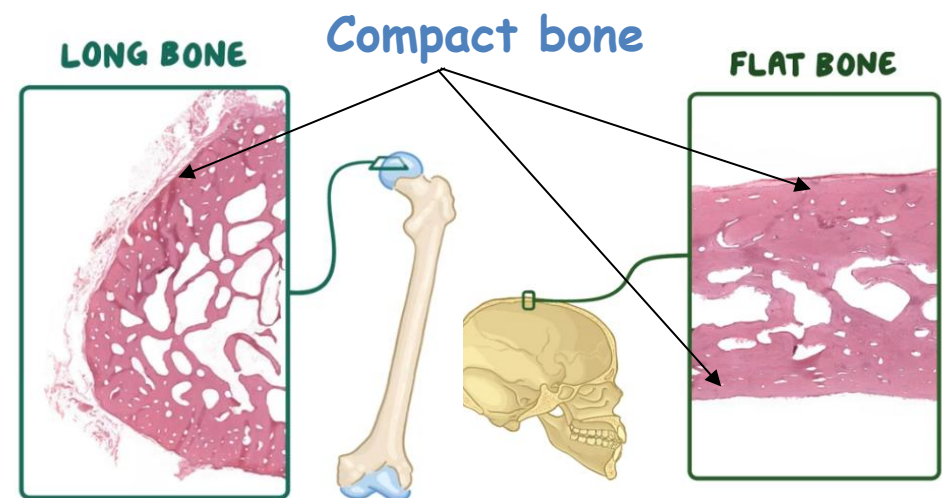
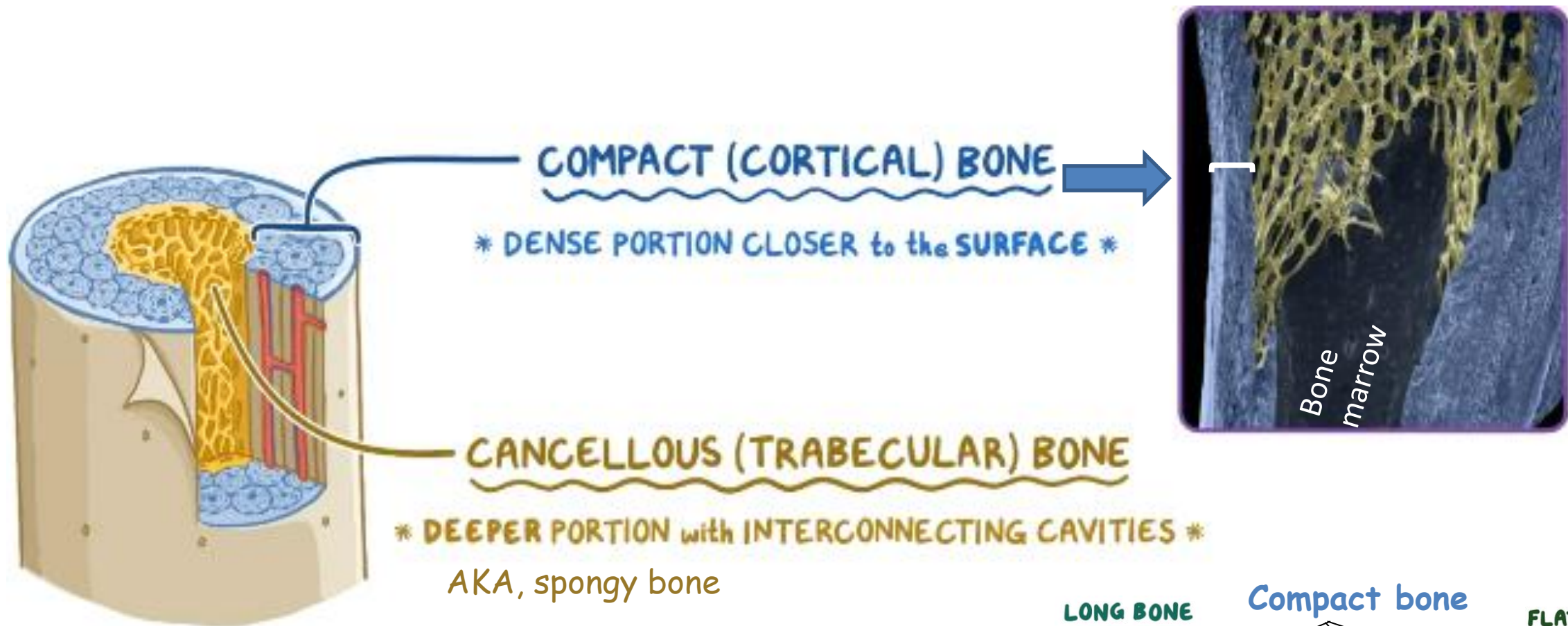
SESAMOID

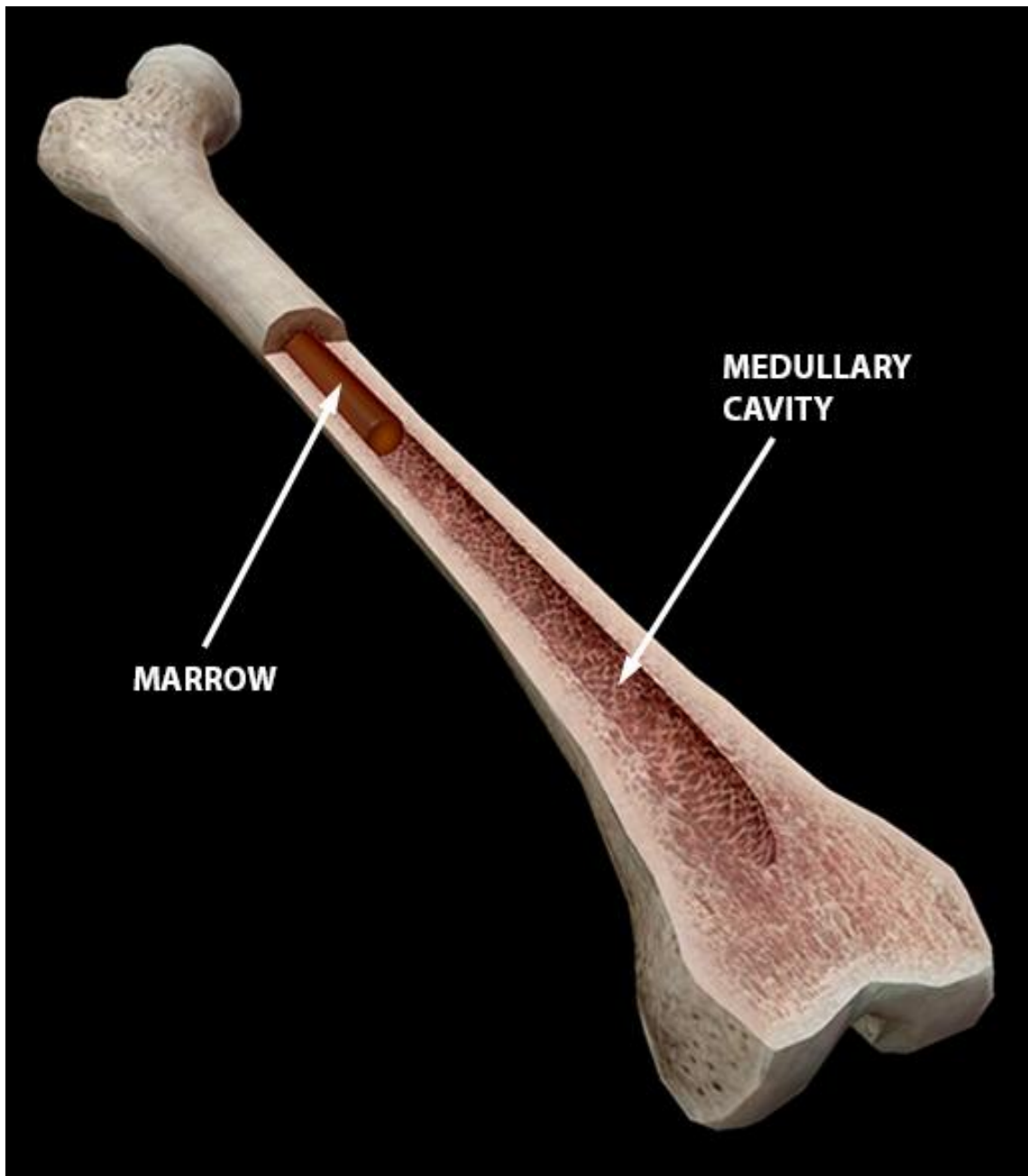


IRREGULAR

Vertebra and
facial bones







(a) Longitudinally sectioned femur (thigh bone)

LONG BONE

FLAT BONE

COMPACT BONE

TRABECULAR BONE

means LITTLE BEAM

SPICULES

~ forms spa

Bone Structure

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graph TD; A[Bone Structure] --> B[Bone Matrix]; A --> C[Bone Cells];
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Bone Matrix

Bone Cells

Type I collagen framework infused with hydroxyapatite crystals $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ + cells (osteocytes, osteoblasts and osteoclasts)

Bone matrix

1) Inorganic materials (60 %; hydroxyapatite crystals).

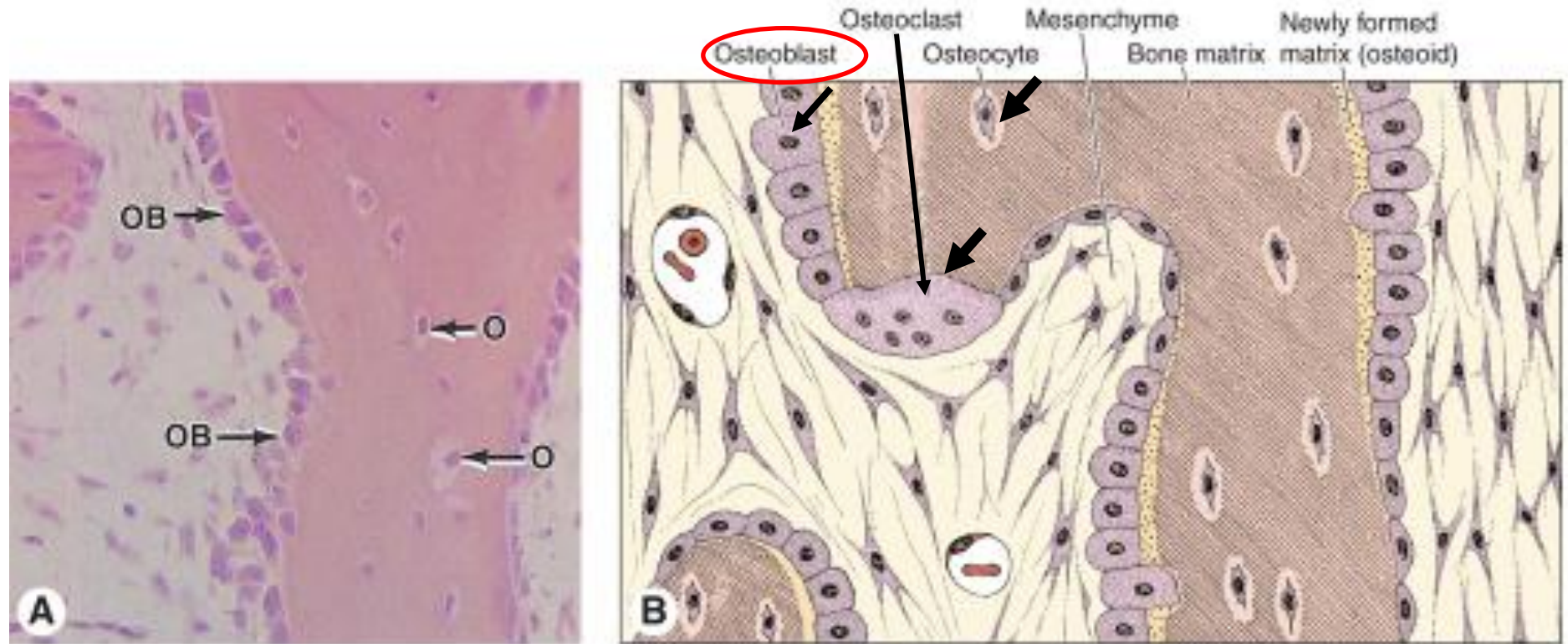
Mostly calcium and phosphate with small amount of bicarbonate, citrate, magnesium, potassium, and sodium ions.

2) Organic matter (40 %; aka Osteoid). Composed of:

- a) fibers (type I collagen, 90%) arranged in an overlapping manner and embedded in unmineralized ground substance.
- b) Ground substance: proteoglycans and multi-adhesive glycoproteins like Osteonectin. Osteonectin is secreted by osteoblasts. High affinity for calcium and collagen, Osteonectin is vital for bone mineralization.

Cellular components

- 1- Osteoblasts
- 2- Osteocytes
- 3- Osteoclast



A–B. Osteoblasts and Osteocytes. (A) The photomicrograph of developing bone shows the location of osteoblasts (OB) and osteocytes (O). (B) Schematic diagram shows the organization of osteoblasts, osteoclasts and osteoblasts within the bone matrix. [Reproduced from Mescher AL: Junqueira's Basic Histology Text and Atlas, 12th edition, McGraw-Hill, 2010.]

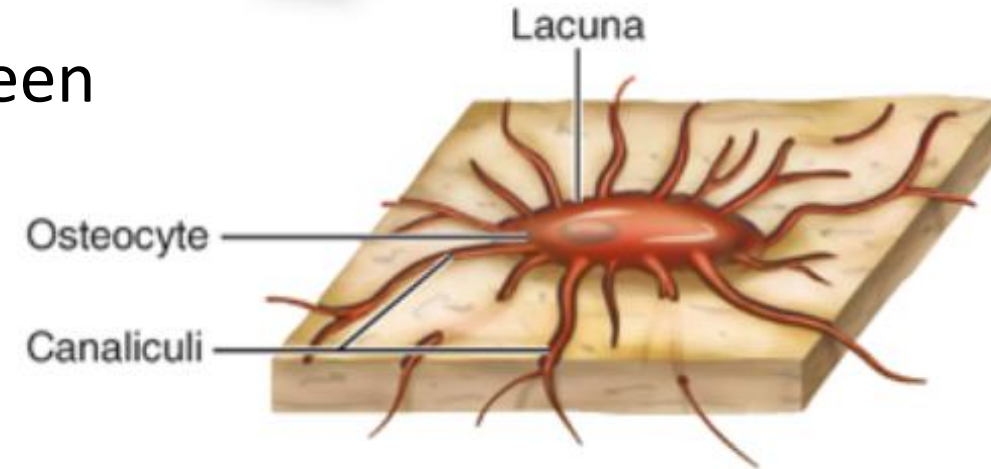
Osteoblasts

- Osteoblasts arise from osteoprogenitor cells under the influence of growth factors (osteoblastogenesis).
- Main bone-forming cells. Cuboidal with one large, round nucleus.
- The osteoblast produce: osteoid protein and chondroitin sulphate (for cartilage) as well as osteonectin.
- Newly formed osteoblasts promote the growth and differentiation of osteoclasts via M-CSF, to maintain the balance between bone formation and resorption.
- Older osteoblasts undergo apoptosis and small population of the osteoblasts differentiate to osteocytes.



Osteocytes

- From osteoblasts.
- Osteocytes are found in cavities (lacunae) between bone matrix layers (lamellae). They are the most abundant cells in bone.
- Osteocytes can contact other osteocytes via a network of canaliculi.
 - Osteocytes are involved in bone turnover.
 - The osteocyte network can be involved in ion exchange, nutrient and waste exchange.
 - Osteocytes can play a role in functional adaptation of bone to maintain homeostasis by regulating calcium flow in or out of the bone and express proteins that can regulate bone remodeling.



[Bone | Junqueira's Basic Histology Text and Atlas, 16e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

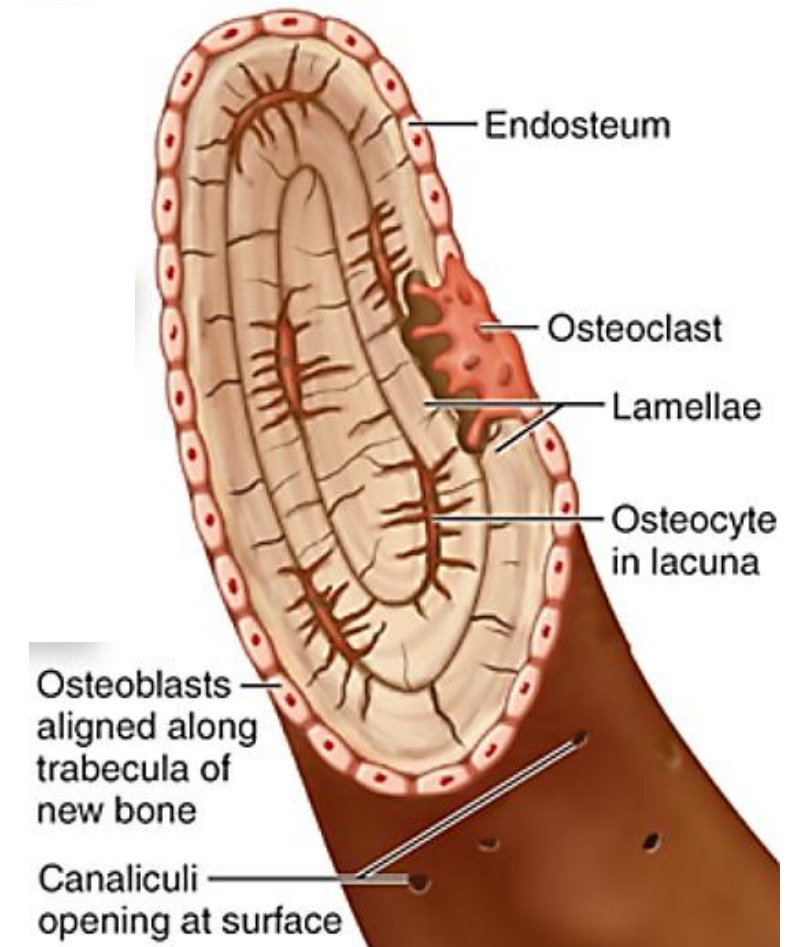
Osteoclasts

- Osteoclasts are large multinucleated motile cells.
- From the bone marrow-derived monocytes.
- Osteoclasts are essential for the resorption and/or remodeling of the bone.
- Both RANKL and M-CSF are required for osteoclast production. Differentiation of osteoclasts is inhibited by the osteoprotegerin (OPG).

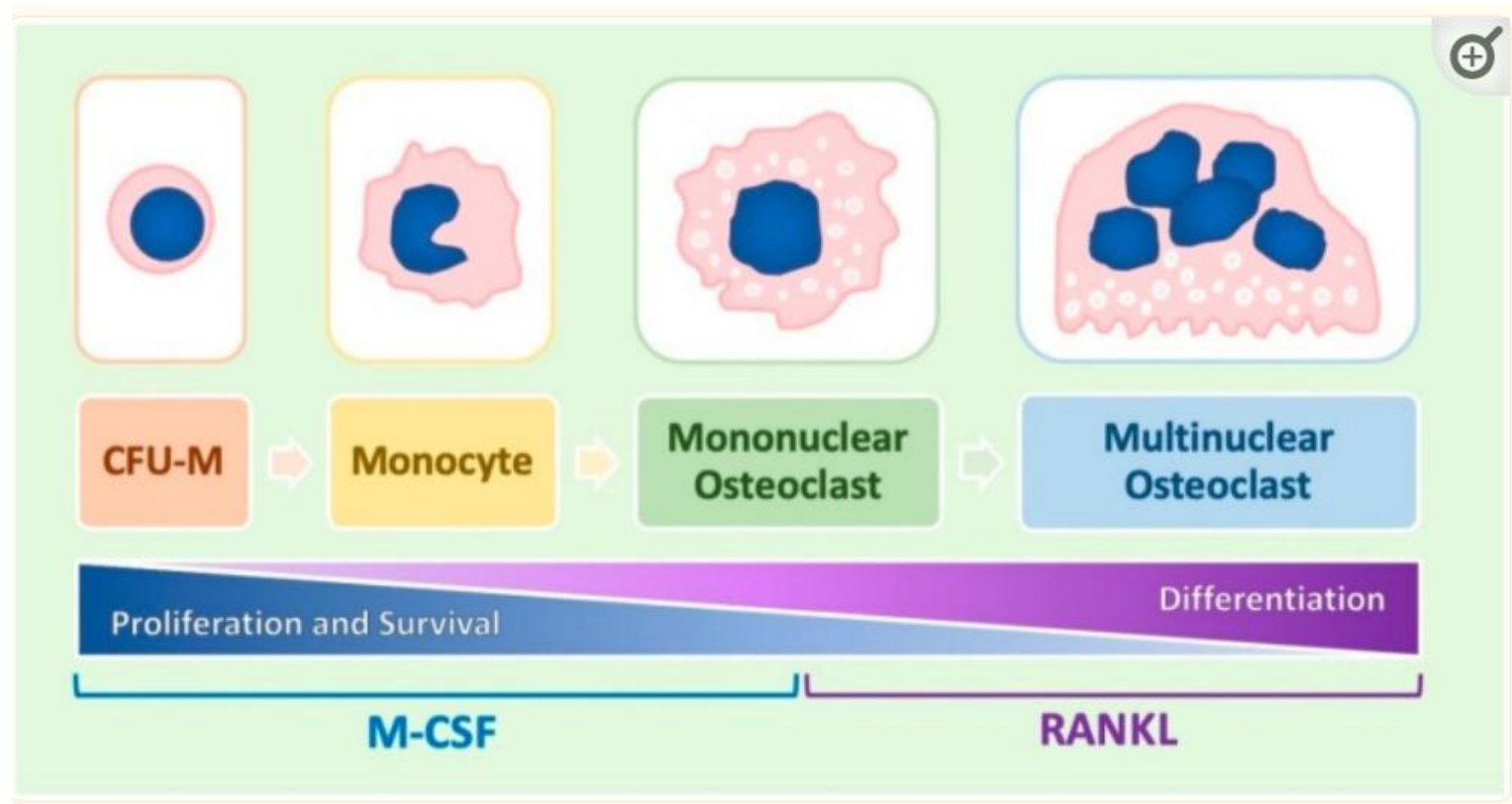
RANK= receptor activator of nuclear factor kappa

RANKL= RANK ligand

M-CSF= monocyte colony stimulating factor.



Osteoclastogenesis



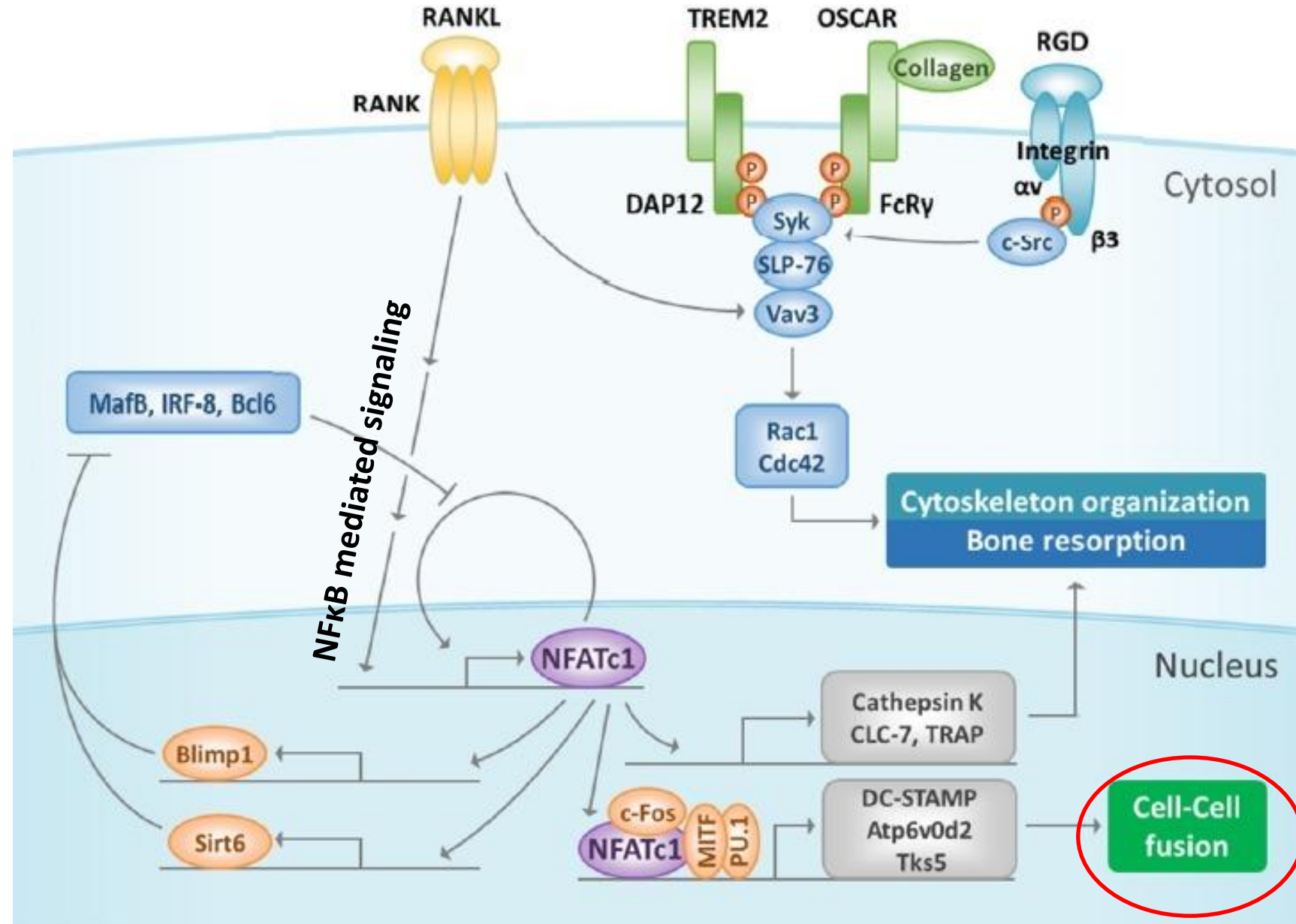
- *Osteoblasts inhibit the formation and activity of osteoclasts via OPG.*
- *Interleukin-6 and tumor necrosis factor can promote osteoclastogenesis. B cells and T cells are the main source of RANKL in periodontitis.*

RANK signaling is mediated via multiple molecules including TRAF6 and NF-κB.

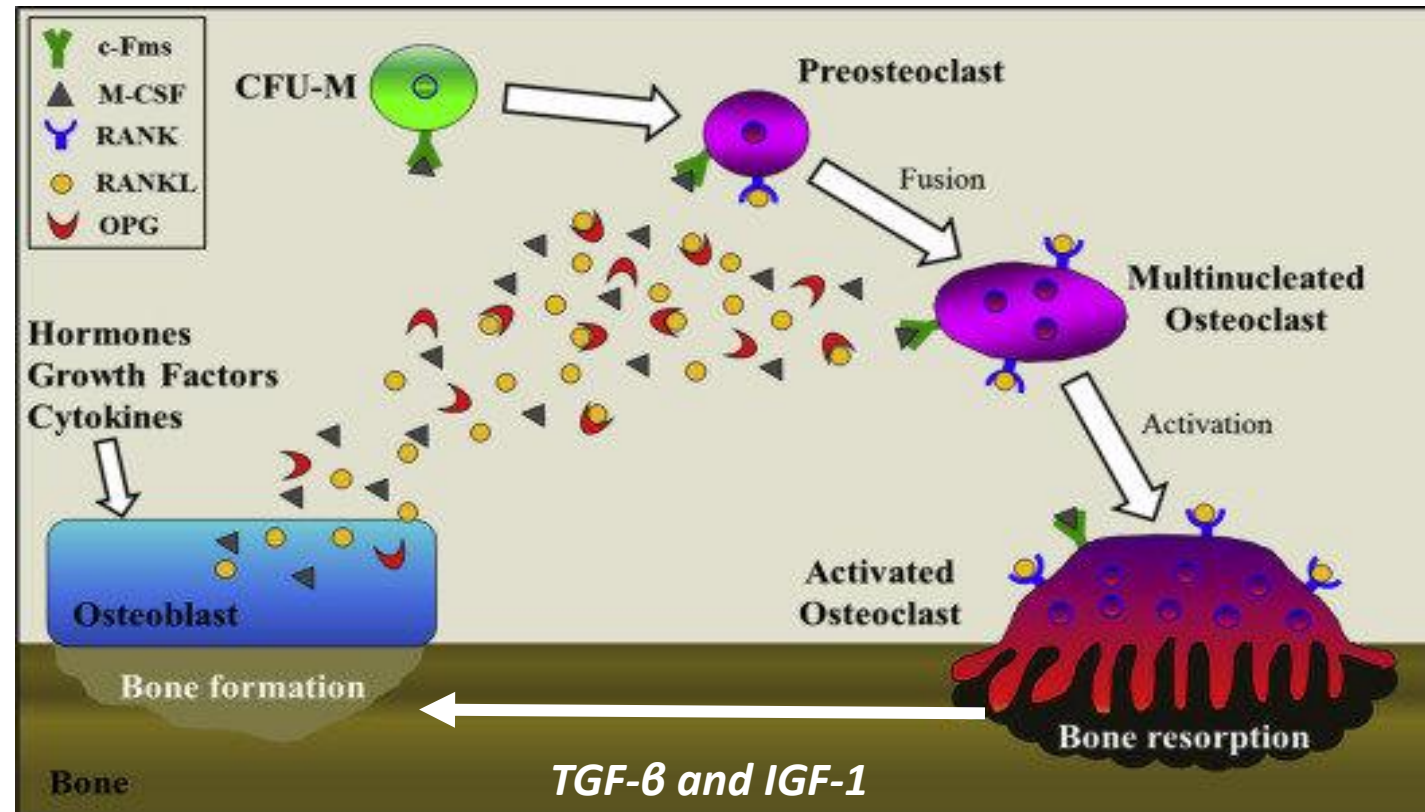
OPG is a decoy receptor for RANKL

Activated NF-κB induces the nuclear factor of activated T-cells cytoplasmic 1 (NFATc1).

NFATc1 induces numerous osteoclast-specific target genes that are responsible for cell fusion and function.

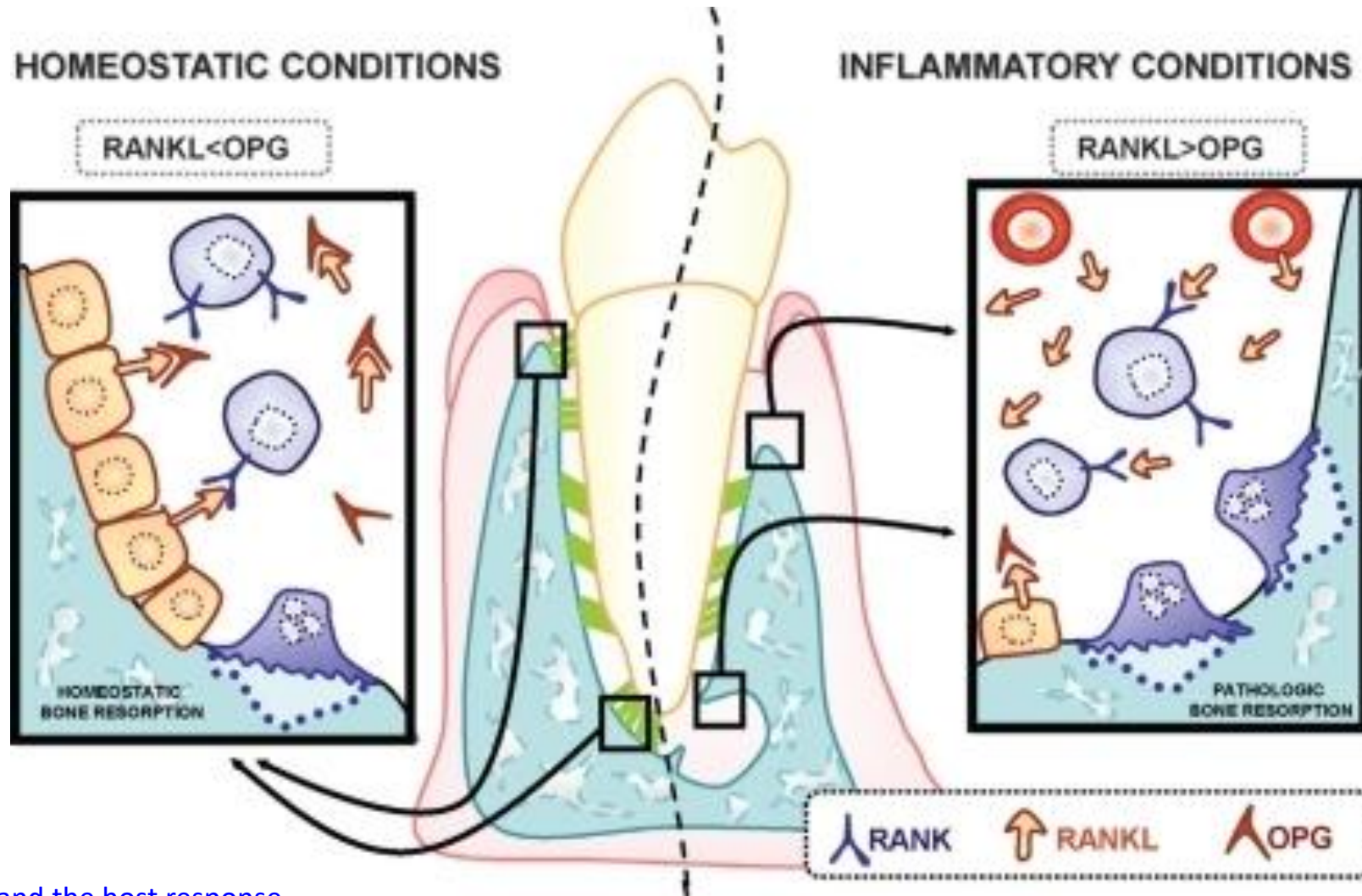


Osteoblasts-Osteoclasts Communication



- (1) Osteoclast-mediated bone resorption releases **TGF-β and IGF-1** from bone matrix to induce osteoblast-mediated bone formation. Osteoclasts secrete other mediators to promote or suppress osteoblast differentiation.
- (2) Osteoblasts secrete **M-CSF, RANKL, WNT5A** that promote osteoclast development as well as **OPG** and **WNT16** to inhibit osteoclast activity.

RANKL/OPG balance is important for bone health



Bone Development and remodeling

- * SPONGY BONE REPLACED EVERY 3 to 4 YEARS
- * COMPACT BONE REPLACED EVERY 10 YEARS



Osteogenesis

Osteogenesis refer to the bone development. It occurs by two main mechanisms:

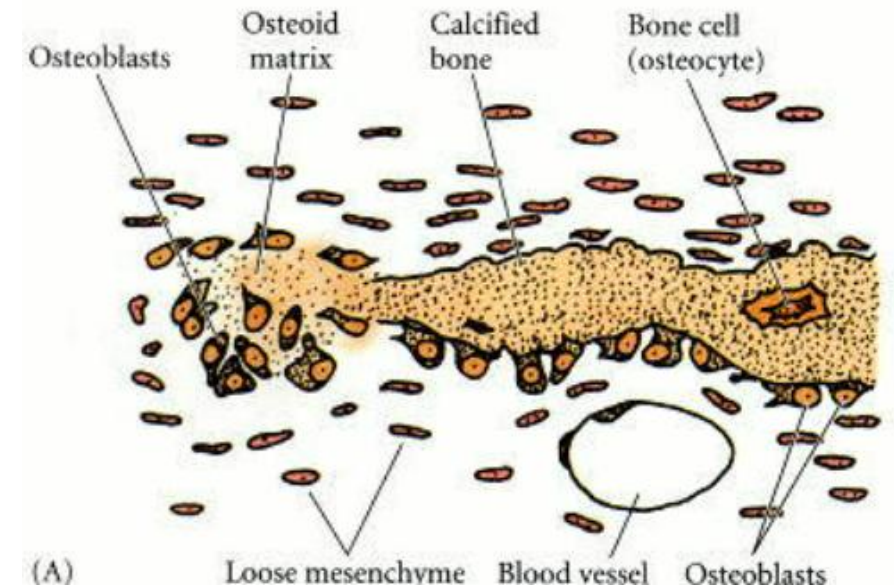
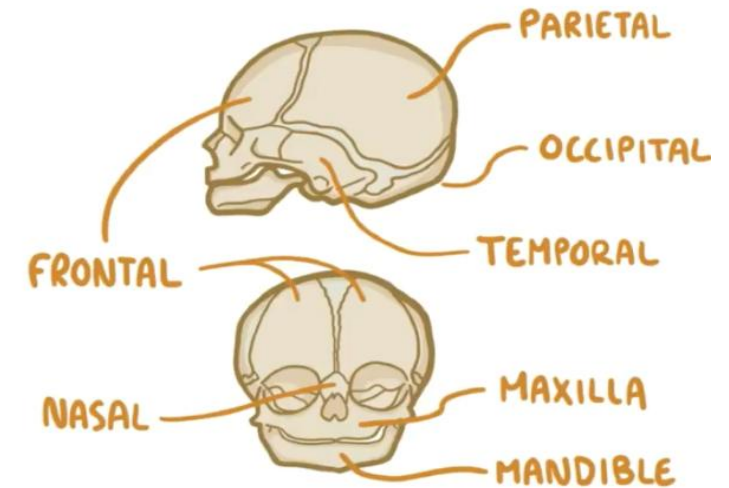
1- Intramembranous ossification, in which osteoblasts differentiate directly from osteoprogenitor cells and begin secreting osteoid. Flat bones, irregular bones

2- Endochondral ossification, occur in a preexisting matrix of *hyaline cartilage* which is invaded by osteoblasts. Then, they begin the osteoid production.

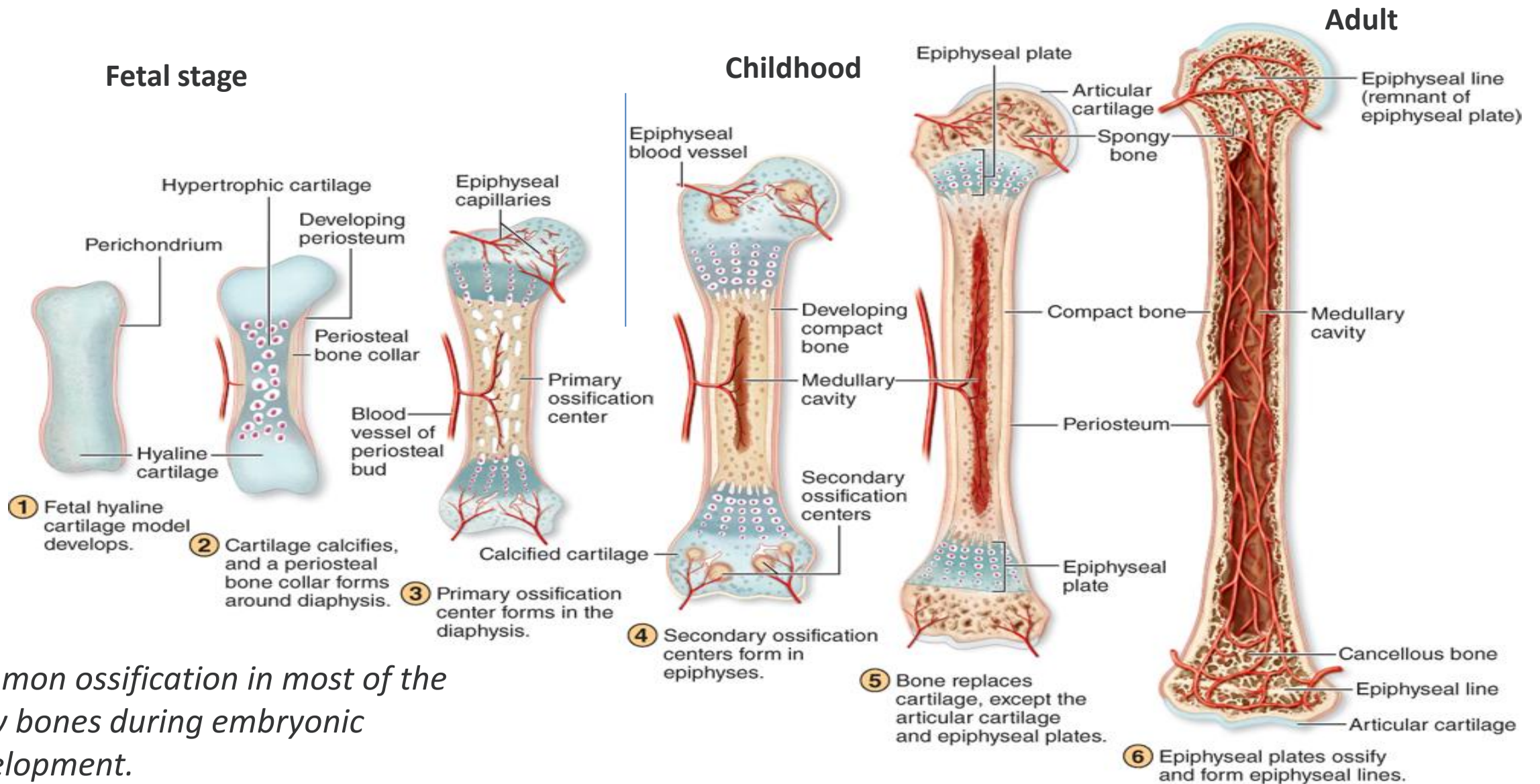
Intramembranous Ossification

- 1- Mesenchymal cells cluster and develop into osteoblasts leading to the formation of a primary ossification center.
- 2- Osteoblasts deposit osteoid matrix. The matrix binds to calcium salts and becomes calcified.
- 3- Continued matrix secretion and calcification enlarges the calcified areas and fuses with nearby ossification centers, leading to the formation of bone without a cartilage backbone.

In some cases, osteoblasts trapped in the calcified matrix develop into osteocytes.

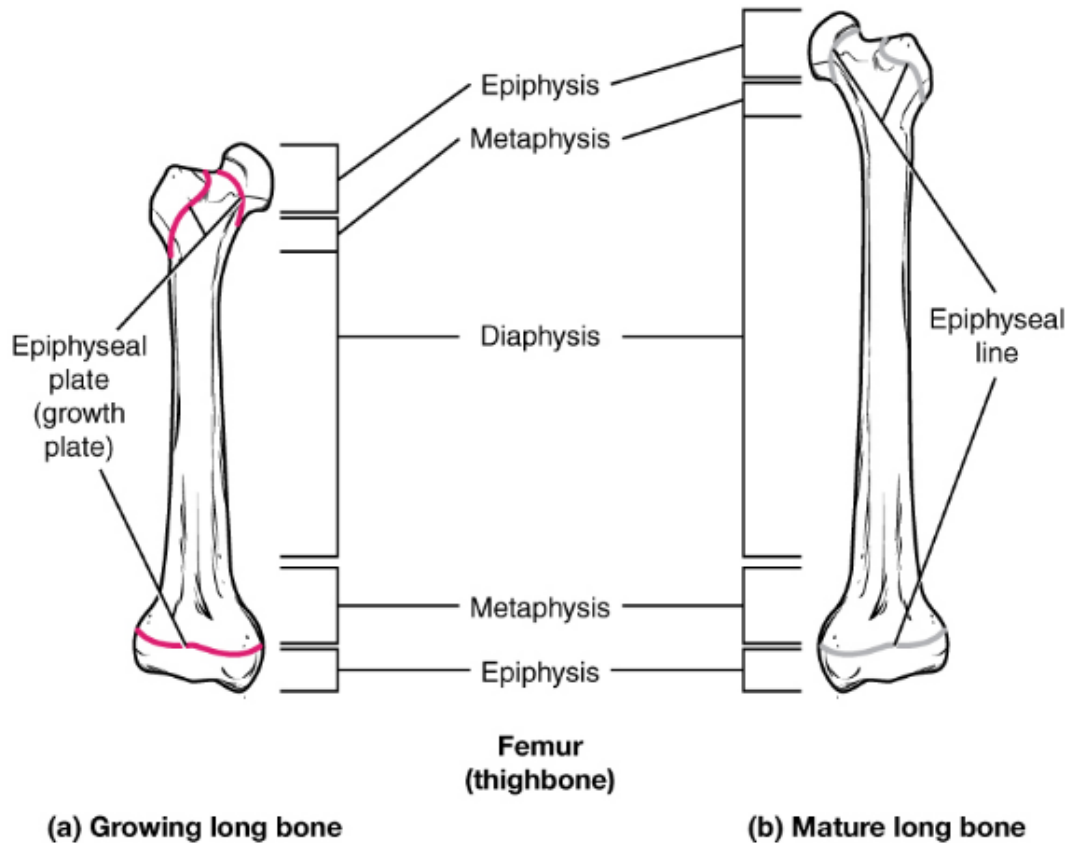


Endochondral ossification

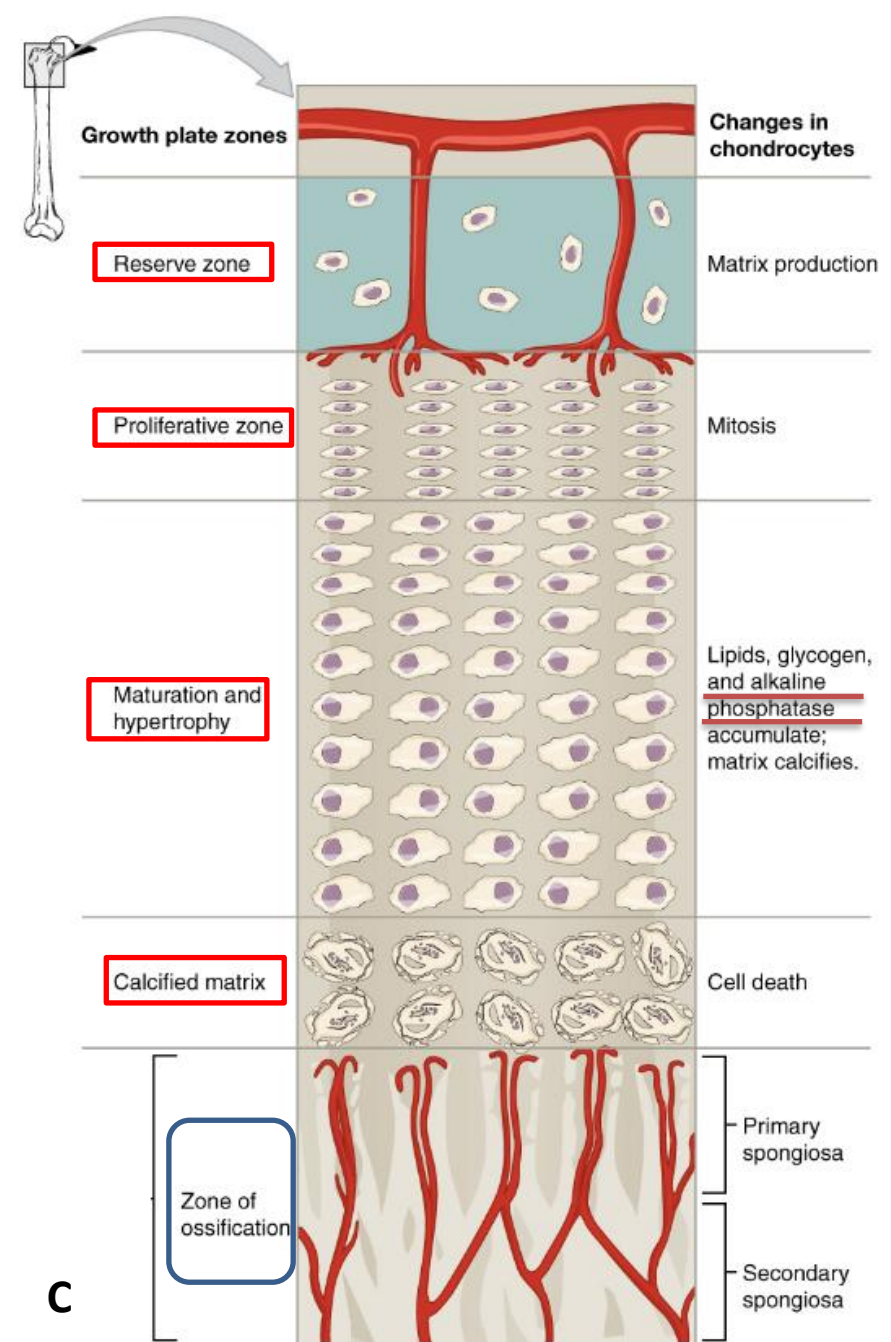


Common ossification in most of the body bones during embryonic development.

Longitudinal Bone Growth



In a mature bone, the thick epiphyseal plate is absent and is replaced by a thin epiphyseal line

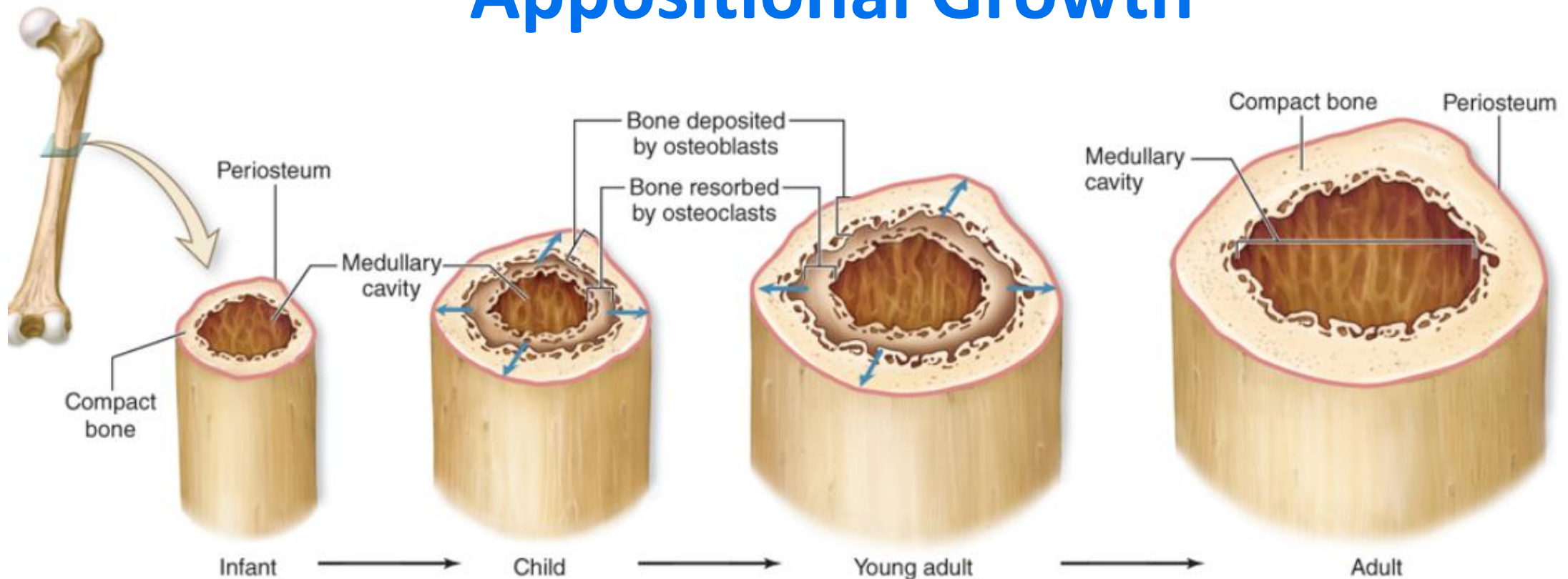


C

The epiphyseal plate is responsible for longitudinal bone growth. The epiphyseal plate is composed of four zones:

- 1- **The reserve zone:** Hyaline cartilage layer with chondrocytes.
- 2- **The proliferative zone**, next layer toward the diaphysis. It contains stacks of slightly larger chondrocyte that are rapidly dividing via mitosis.
- 3- **Zone of maturation and hypertrophy.** In this layer, chondrocytes are undergoing final maturation, and grow in size (hypertrophy). In this layer, the chondrocytes secrete type X collagen, and the calcification of the matrix begins leading to an eventual chondrocyte death.
- 4- **Zone of calcified matrix**, zone closest to the diaphysis. Chondrocytes are dead because the matrix around them is calcified. Capillaries and osteoblasts from the diaphysis penetrate this zone. The osteoblasts start the ossification process by the formation of hydroxyapatite crystals on the calcified cartilage matrix. The bone grows in length when osseous tissue is added to the diaphysis. Bones continue to grow in length until early adulthood. The longitudinal growth stops when chondrocytes in the epiphyseal plate stop proliferating and bone replaces the cartilage.

Appositional Growth



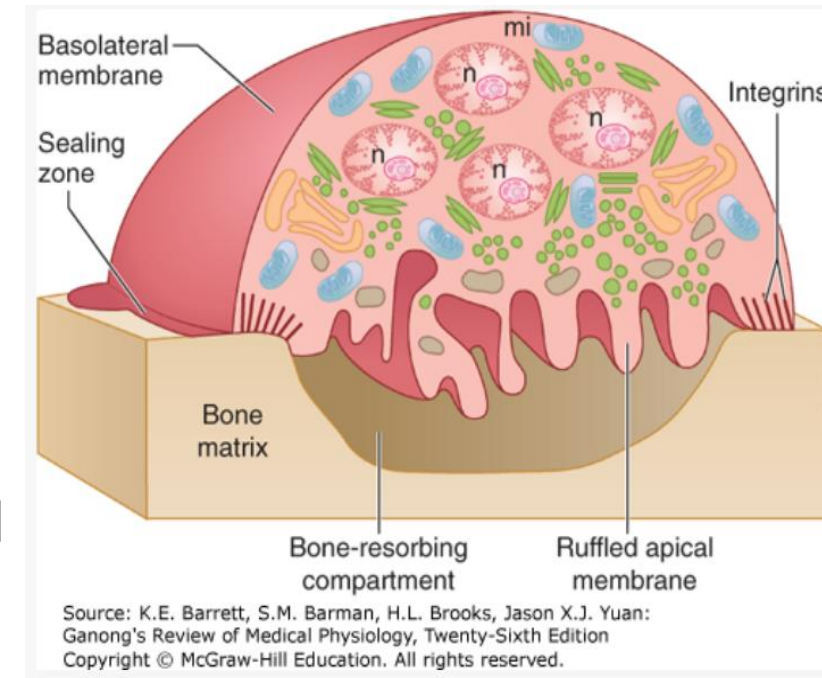
Source: Anthony L. Mescher:
Junqueira's Basic Histology Text and Atlas, 16e

Appositional growth: Growth in the circumference through osteoblasts (osteoprogenitor in the periosteum) forming new bone tissue and osteoclast resorbing old bone in the medullary cavity.

Bone remodeling: bone resorption

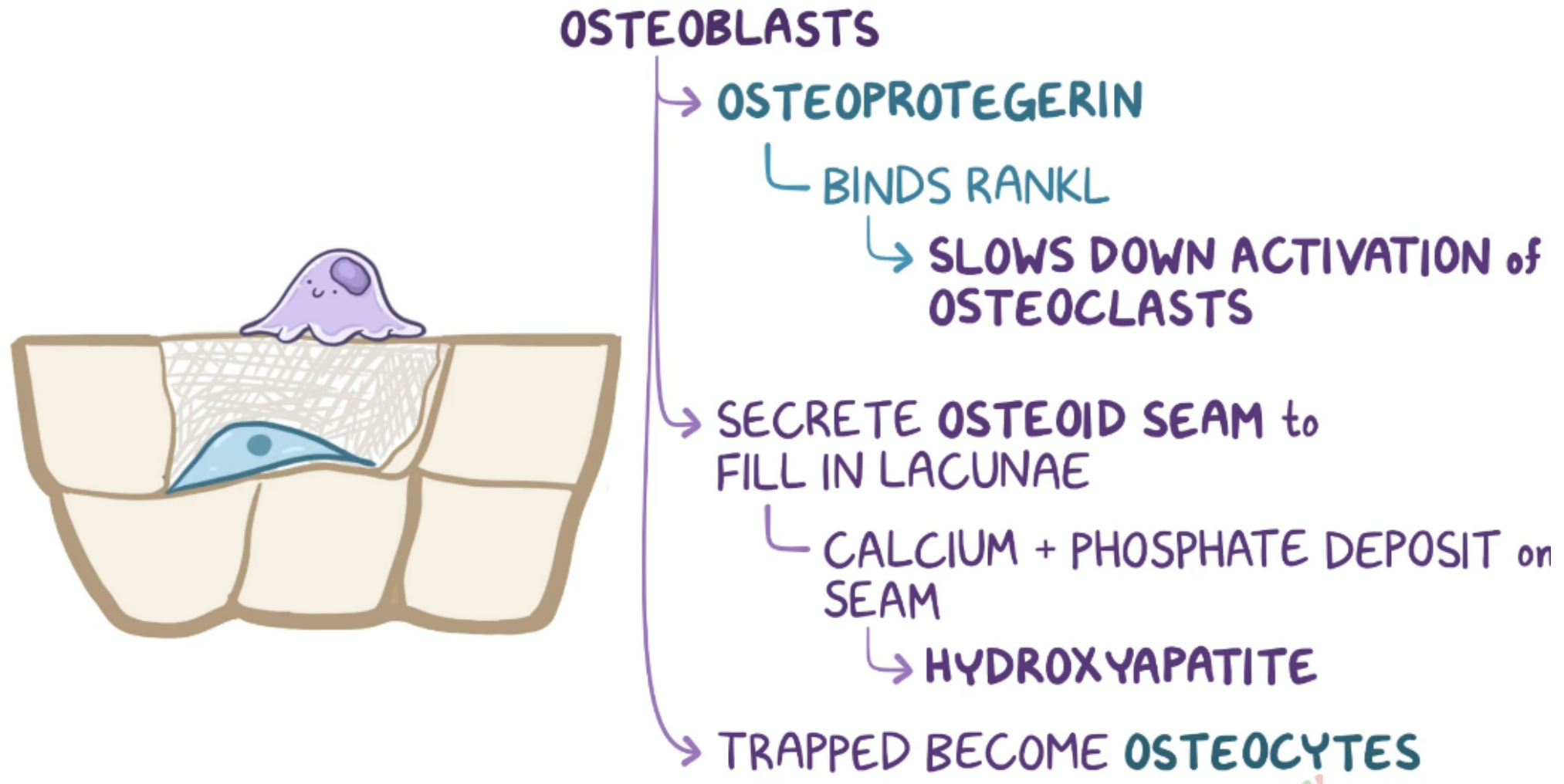
Osteoclasts erode and absorb previously formed bones.

- Sealing zone: the edge of osteoclast are tightly sealed to the bone via integrins to create a bone-resorbing compartment.
- Osteoclasts secrete lysosomal enzymes (mostly collagenase) to break down the collagen matrix, forming shallow depression in the bone (Howship's lacunae).
- Osteoclasts also secrete hydrochloric acid via a proton pump to acidify the area to around pH= 4.0. The acidic pH dissolves hydroxyapatite into soluble calcium and phosphate.
- Metabolized bone digestion products move to the blood stream (calcium and phosphate)



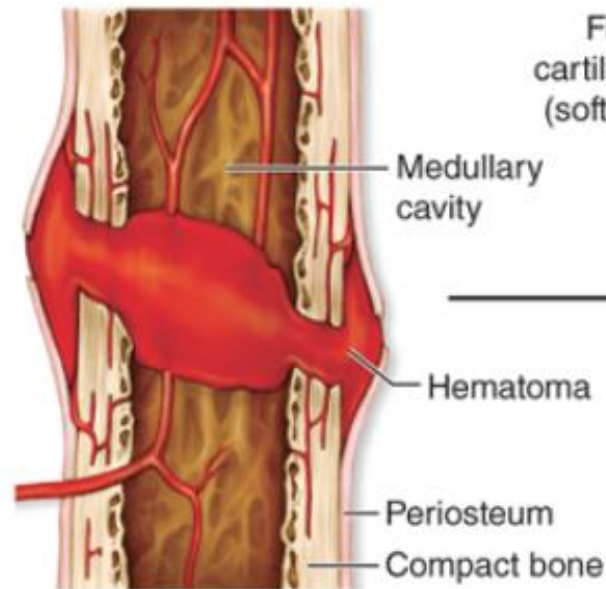
[Hormonal Control of Calcium & Phosphate Metabolism & the Physiology of Bone | Ganong's Review of Medical Physiology, 26e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

Bone remodeling: Bone deposition

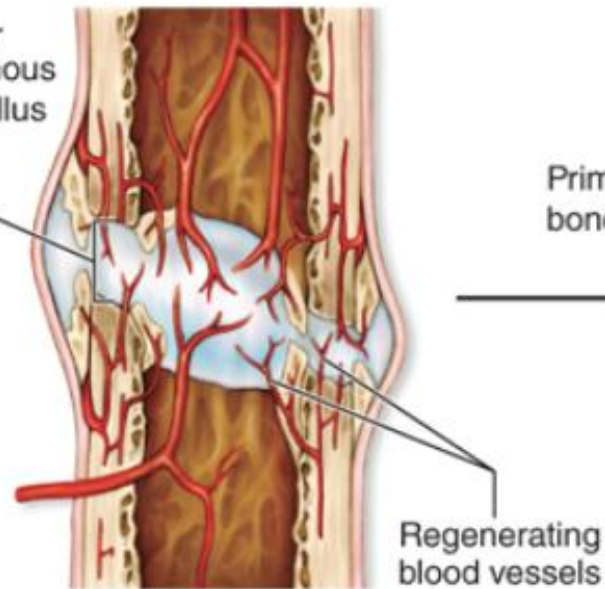


Bone Repair

Reactive phase

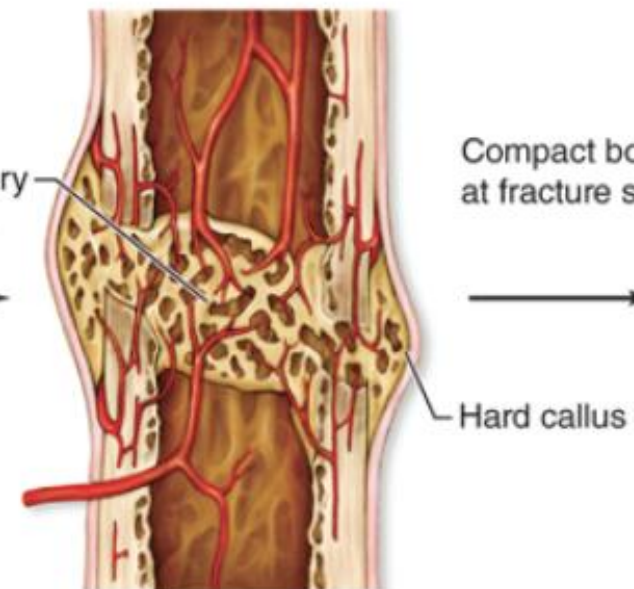


① A fracture hematoma forms.



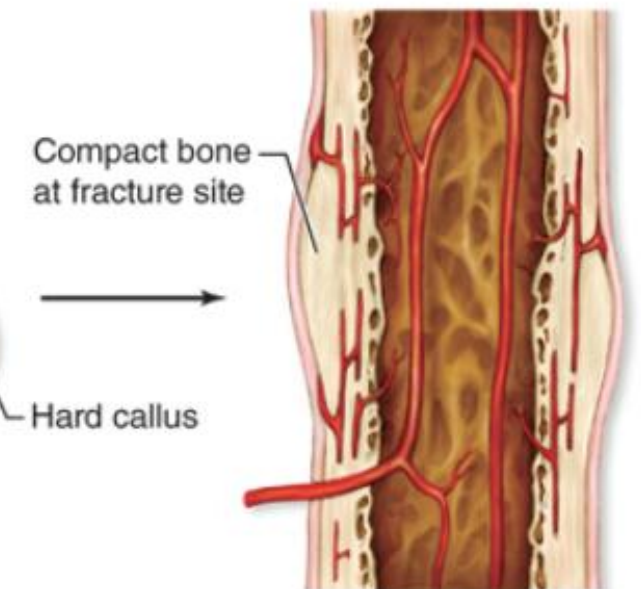
② A fibrocartilaginous (soft) callus forms.

Reparative phase



③ A hard (bony) callus forms.

Remodeling phase



④ The bone is remodeled.

Bone remodeling in response to normal growth, microfractures or trauma-induced fractures is tightly regulated process.

1- Reactive phase.

Immediately after a fracture with blood vessel constriction and clot formation. This hematoma is replaced by a fibrous connective tissue (soft callus or procallus) tissue with regenerated blood vessels. This will also initiate an inflammatory reaction and stimulate the release of growth factors.

2- Reparative phase.

Within few days after the fracture, chondroblasts and osteoblasts develop within the periosteum. These cells produce an initial woven bone, named “callus,” which unites the broken bone fragments and provides support for the fracture. The hard callus is formed of hyaline cartilage (type II collagen). Next, this hyaline cartilage matrix is replaced by a lamellar bone (type I collagen) via ossification.

3- Remodeling phase:

Remodeling can take from several weeks to months to complete. It includes bone resorption by osteoclasts and redeposition by osteoblasts until a final compact bone matrix is produced, closely mimicking the original bone.

Regulation of Bone remodeling

Bones contain approximately 99% of the body's total calcium. Regulation of calcium levels in the body influence the balance of bone formation and breakdown.

1- Parathyroid gland

- Via PTH (peptide hormone). Osteoblasts has the PTH receptors while osteoclast has no receptors for PTH.
- Low serum calcium stimulates PTH release. High PTH causes catabolic effects on the bone.
- PTH raises low blood calcium levels by stimulating the osteoblasts to release RANKL. RANKL triggers osteoclasts, bone resorption and release of calcium ions into the bloodstream.
- In contrast, PTH is suppressed in higher-than-normal calcium level. Low PTH has anabolic effects on the bone.

2- Thyroid gland

- High calcium level stimulates calcitonin from the thyroid parafollicular cells. Calcitonin binds to osteoclasts & inhibit bone resorption to decrease the elevated blood calcium levels

3- Estrogen

- can slow bone resorption by inhibiting the production of bone-eroding cytokines. Inhibits osteoblast apoptosis and promote osteoclast apoptosis.

4- Growth hormone

- From pituitary. Enhance the bone growth.
- Lack of growth hormone during childhood causes pituitary dwarfism.
- Excessive production of growth hormone in childhood can cause gigantism.

5- Mechanical stress can increase the rate of bone remodeling.

- Bones that bear a lot of weight remodel at a higher rate (Wolff's law).

6- Active vitamin D

- stimulates calcium absorption from the intestine which increases the calcitonin level and inhibit bone resorption.

7- Vitamin C is required for collagen production.

Markers for bone disorders

- PTH: Increases in hyperparathyroidism and it affect the calcium levels (increase).
- Serum calcium and phosphate levels: In general, there is an inverse relation between Ca^{+2} and PO_4^- . Calcium increase, while phosphate decreases in primary hyperparathyroidism.
- Alkaline phosphatase (bone-specific ALP isoform): found on osteoblast and level is reflective of bone mineralization and formation.
- Active vit. D level (calcitriol).

Bone, Enamel, Dentin and Cementum

- Enamel is the most brittle, 95 % mineral, 2% matrix protein and 3 % water.
- Dentin and cementum resemble the compact bone. Bone, dentin and cementum are around 70 % mineral, 20 % collagen and 10 % water.
- Enamel matrix is partially mineralized while developing. While, bone, dentin and cementum are secreted unmineralized, then calcium mineralizes and promote hydroxyapatite crystals formation.

Resources

